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Research paper

Integrated audit intelligence: A multi-task learning model for classification and opinion generation in power systems

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ABSTRACT

Auditing in electric power systems involves complex decision-making based on large volumes of unstructured and domain-specific records. Under the digitalization and decarbonization of integrated energy operations, such audit records also reflect cyber-physical and operational risks, so Artificial Intelligence (AI)-driven audit intelligence can support resilience enhancement by improving risk visibility and accelerating standardized corrective decision-making. Automating this process requires models that can understand context, identify issues, and generate relevant textual outputs. This research presents an integrated Multi-Task Learning framework for power-system auditing that jointly performs audit issue classification and audit opinion generation within a unified model. The system employs multiple encoders and decoders anchored on a pre-trained transformer-based language model to process heterogeneous audit content and learn both common and task-specific patterns. Experiments on real-world datasets from electric utilities demonstrate that the proposed multi-task learning framework outperforms conventional single-task baselines in both classification and generation tasks. The model delivers consistent improvements in classification metrics such as F1-score (harmonic mean of precision and recall), along with enhanced generation quality across Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Distinct-n (lexical diversity), and Bidirectional Encoder Representations from Transformers Score (BERTScore), which measures semantic similarity. Additional ablation studies are conducted to assess the contribution of different input types. The results confirm the effectiveness of the proposed approach and highlight its potential as a scalable and intelligent solution for advancing audit automation in the power sector.

1. Introduction

Auditing plays a vital role in the management of electric power systems and directly influences the financial integrity, regulatory compliance, and operational reliability of power enterprises (Deng et al., 2024). Recent risk-based optimization for maintenance and replacement of substation equipment further highlights the operational consequences of audit findings and motivates more data-driven audit intelligence in power enterprises (Zhou et al., 2025). This also aligns with the focus on Artificial Intelligence (AI)-driven resilience enhancement in integrated energy systems, where trustworthy decision support is needed to manage complex cyber-physical and operational risks in real-world settings (Farina and Pedrycz, 2024). Moreover, uncertainty-aware operational analysis in power systems, such as interval power flow

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under bounded hybrid uncertain sets, underscores the need for risk-aware decision support in modern power and integrated energy operations (Wang et al., 2024a). In parallel, AI-driven cyber-resilience enhancement in integrated energy system scheduling using demand response and deep reinforcement learning provides additional motivation for trustworthy analytics that connect enterprise governance with resilient operation (Li et al., 2025a). During the audit process, auditors are typically responsible for completing four key data fields within an audit form: the audit description, referenced provision, audit issue category, and audit opinion, as outlined in Table 1. The audit description summarizes the auditing procedure and its findings, while the referenced provision specifies the regulatory or policy-based criteria used to evaluate compliance. Both of these elements are manually prepared by auditors based on observed activities and documented evidence. Following this, auditors assign a categorical label to the audit issue to facilitate standardization, and ultimately, they formulate an audit opinion to reflect the overall outcome of the review.

While these tasks are conventionally performed manually, advancements in deep learning techniques have demonstrated potential for automating certain components, particularly audit issue classification and opinion generation. Audit issue classification can be modeled as a standard multi-class text classification problem, where the goal is to assign the most appropriate category to a given textual audit record (Deng et al., 2024). Prior studies in the power domain have employed word embeddings, statistical models, and more recently, deep neural networks (Meng et al., 2023) to achieve this. Similarly, audit opinion generation can be framed within the broader field of Natural Language Generation (NLG), where the aim is to generate concise, meaningful summaries based on textual inputs. This task shares similarities with applications such as dialogue generation, summarization, and machine translation, all of which have been widely explored using Sequence-to-Sequence (Seq2Seq) architectures powered by pre-trained language models (Kelley and Hagan, 2025), (Wu et al., 2025).

In the case of audit data, the audit opinion serves as a summary based on the audit description and referenced provision. As illustrated in Table 1, there is a clear lexical and semantic alignment between these fields, which supports the feasibility of adopting a Seq2Seq framework for generating audit opinions. Existing work has shown that audit issue classification and audit opinion generation can each be addressed individually with satisfactory results. However, treating them as independent tasks overlooks the potential benefits of their intrinsic relationship. In practice, understanding the audit issue category often provides strong contextual cues for drafting an appropriate opinion. Conversely, the structure and content of a well-written audit opinion may help inform the classification of the issue. This interdependence suggests that the two tasks can reinforce one another if learned jointly under a multi-task paradigm.

In this work, we propose a unified multi-task learning framework that simultaneously performs audit issue classification and opinion generation. To achieve this, we develop a dual-encoder architecture, utilizing a pre-trained Bidirectional and Auto-Regressive Transformers (BART) model to encode audit descriptions (Soper et al., 2021) and a domain-specific legal encoder to process the referenced provisions (Xiao et al., 2021). The encoded features are fused through a self-attention mechanism and passed to both a classification head and a decoder for generation. For deployment in real-world audit infrastructures, such models can be complemented with secure distributed communication protocols (He and Choo, 29, 2022) and privacy-preserving federated learning with verifiable trust-aware aggregation, as reported for wind power data imputation in a zero-trust setting (Li et al., 2025b). Thecategory of the audit issue is forecast through a fully connected layer, and the audit opinion is produced through a BART-based decoder. A composite loss function is proposed to optimize both the classification and generation tasks jointly, and the model is trained with the Adaptive Moment Estimation with Weight Decay (AdamW) optimizer (Vandenhende et al., 2021).

Table 1
Example audit records used in this study, including issue category, audit description, referenced provision, and audit opinion.

Issue Category	Problem Description	Referenced Provision	Audit Opinion
Inaccurate billing audit	The project settlement approved the completion surveying and mapping fee of 76,230 CNY (71,915.09 CNY excluding tax), approved 76,230 CNY (74,009.71 CNY excluding tax), miscalculated the contract value-added tax rate from 3% to 6%, and the settlement audit was not strict.	"Administrative Measures for the Settlement of Power Transmission and Transformation Projects of ***** Company": "The settlement price of the project shall be agreed in the following ways: (1) The winning bid price and Part of the amount can be adjusted according to the contract"	Strengthen the project settlement review work, review the relevant costs in accordance with the contract, and improve the accuracy of the project settlement.
Project accounting adjustment is not timely	In these companies, 180 project accounting adjustments were not timely, and the project accounting adjustments were not completed within 30 days after the final accounts were approved, involving 1233 accounting adjustments and an amount of 136,696,400 CNY.	"Measures for the Financial Management of Projects of ****** Company": "The legal entity of the project shall complete the relevant accounting adjustment within one month according to the approval of the final accounts."	Relevant units should strictly check the authenticity and rationality of the results report to ensure the authenticity, accuracy and completeness of the project data.
Error in estimation of design cost	(cd-marketing 18–04) For 3 projects including the new charging pile group in Xiaodian Town, District BC, the estimated design cost is calculated according to the "overhead line and cable line project", and the actual should be based on the "distribution station, switch station and charging (Replacement) Power Plant Project", an artificially high 236,600 CNY.	"Administrative Measures for Marketing Projects of **** Company" "The main contents of the preliminary design review: … (8) Verifying the basis for preparing the estimate, the depth of the design, and the cost Composition, calculation standards, division of the nature of projects and costs, normative budget tables, etc.; verify the calculation of various costs, whether there are omissions or double counting."	Effectively perform the preliminary design preparation and review responsibilities, and strengthen the whole-process control of project design quality.
Insufficient design depth	From 2016–2020, three companies jz, nh, and wq have completed agricultural power grid transformation and upgrading projects. Among	"Provisions for the Management of 10 (20) kV and Below Distribution Network Engineering Projects of **** Company": "Once the distribution network	Relevant companies should implement the preliminary design review responsibilities, strengthen the depth of engineering design

(continued on next page)

Table 1 (*continued*)

Issue Category	Problem Description	Referenced Provision	Audit Opinion
	them, the estimated budget of 10 projects has changed by more than 40% compared with the settlement, and the preliminary design depth of the project is insufficient, involving a difference of 10.4606 million CNY.	engineering design is approved, it must be strictly Implementation. The procurement and use of equipment and materials should meet the requirements of the design plan, and the construction unit must strictly follow the drawings for construction."	and on-site management, and strengthen the management of the entire process of project cost.

In order to test the proposed method, we built a real-world audit dataset from the power industry audit department and conducted extensive experiments. Our assessment covers a set of metrics for both classification and generation quality. Experimental results show consistently that our multi-task framework performs better than single-task models in both tasks. Further, ablation studies are performed to measure the contribution of each data point to overall model performance and gain insights that can inform future automation and assist manual audit workflows. The versatility of our framework also indicates potential for wider use in other domains of audit, towards enhanced efficiency and lower operational expenses.

The main contributions of this paper are summarized as follows:

- This study formulates an integrated audit-intelligence setting for electric power enterprises that jointly supports audit issue classification and audit opinion generation, providing a system-level multi-task learning application aligned with practical audit workflows.
- This paper develops a dual-encoder and fusion architecture tailored to heterogeneous audit inputs, where BART encodes auditor-written descriptions and LawFormer encodes provision-style legal/technical text, and a self-attention fusion mechanism is used to capture cross-text semantic alignment.
- Extensive experiments and ablation analyses demonstrate that joint multi-task supervision improves both classification performance (macro-averaged F1 score, macro-F1) and generation quality measured by BLEU, ROUGE, and BERTScore, and further reveal how each input component and auxiliary task contributes to the overall gains.

The remainder of this paper is organized as follows: Section 2 reviews related work on power-domain text classification and audit opinion generation. Section 3 presents the proposed dual-encoder multi-task framework, including the fusion mechanism and joint optimization. Section 4 describes the dataset, experimental settings, baselines, and results with extended analyses. Section 5 concludes the paper and discusses implications for practical deployment.

2. Related work

2.1. Text classification

Text classification has gained increasing significance in electric power systems, for which it facilitates a variety of applications ranging from issue detection to document tagging and smart decision-making. These techniques are applied to automate tasks, increase efficiency, and minimize operating costs. With the advent of deep learning, a number of neural network models have been used successfully for this purpose, such as FastText, Long Short-Term Memory (LSTM), and Convolutional Neural Network for Text (TextCNN). More recent research has investigated power-specific approaches. For instance, category-mixed embedding hierarchical classification models have been developed to enhance the structural awareness of electric power texts (Urlana et al., 2022). Previous methods like Term Frequency–Inverse Document Frequency (TF-IDF) and word embedding-based classifiers have also been used in audit text classification tasks (Meng et al., 2023). Domain-specific dictionaries were introduced in some studies, and Bidirectional Long Short-Term Memory (BiLSTM) networks were utilized to extract sequential features in audit texts. Additional improvement came from incorporating attention mechanisms with BiLSTM models, enhancing the capacity to extract meaningful information from power equipment defect records (Yang et al., 2022). Related power-domain intelligence studies have also emphasized lightweight multisource information fusion for fault diagnosis (Hang et al., 2024) and physically interpretable frequency-aware deep models for fault prediction (Wang et al., 2024b), supporting the value of representation learning under heterogeneous and multi-source technical records.

These initiatives have proved the efficacy of deep learning for power-based classification tasks (Deng et al., 2024). In addition, self-supervised condition evaluation pipelines for critical electrical components have been reported, which further increases the volume of technical inspection evidence that can enter enterprise audit and compliance workflows (Cao et al., 2025). Nonetheless, the majority of current studies deal with text classification as a standalone task. In realistic audit situations, classification is sometimes tied to another duty, like creating summaries or writing conclusions. This research draws on previous work by adding another but connected task, audit opinion creation. Using a multi-task learning model, common context among these tasks is used to enhance their individual performance.

2.2. Natural language generation

NLG is a central subfield of NLP that deals with producing coherent and meaningful text from input provided. Among the earliest and most popular methods is the Seq2Seq model (Kelley and Hagan, 2025). It takes an input sequence and produces an output sequence token by token until a response is fully created. Seq2Seq model has been successfully utilized in various tasks like machine translation, where the model translates text from a source language to another target language, and in dialogue systems, text summarization, etc.

Pre-trained language models have been developed in recent times, and these have improved text generation task performance immensely (Song et al., 2023). Among them, the BART model is most effective, providing a strong Seq2Seq structure based on the Transformer network (Wu et al., 2025), (Naseem et al., 2021). BART can generate various generation tasks with high accuracy and flexibility. In this research, we consider audit opinion generation as a Seq2Seq task. The input is the audit description and the provision referred to, while the output is the final audit opinion.

While audit opinion generation in the electric power industry has not been widely researched, comparable work in proximal areas supports its viability. For example, generative models have been employed with success by some to generate structured summaries of scientific papers, comparing favorably to classical deep learning baselines (Hofstätter et al., 2022). Other work has shown the applicability of Transformer-based models to code generation and summarization, tasks with a similar structure and complexity (Bujnowski et al., 2020). For example, recent advancements such as SecureGPT showcase how Transformer models can be extended to privacy-sensitive domains (Zeng et al., 2024). Such work justifies the applicability of pre-trained models for audit opinion generation in electric power auditing systems.

2.3. Multi-task learning

Supervised learning is a popular paradigm of machine learning, that

encompasses a vast number of tasks such as classification, regression, generation, and ranking. Still, training a model on one task tends to produce suboptimal results for lack of contextual learning. In most instances, information available in similar tasks is not utilized. multi-task learning (Urlana et al., 2022) provides a solution to this shortcoming by allowing a model to be trained on several tasks at once so that shared representations can fill informational gaps. In the power and energy domain, multi-task end-to-end long-sequence modeling has also been studied for load forecasting, demonstrating that shared representations can improve performance across correlated targets (Zhang et al., 2025). The value of interpretable shared learning in multi-context decision problems has been further reinforced in fuzzy clustering research (Alateeq and Pedrycz, 2024). Beyond multi-task structures, interpretable learning has also been advanced through higher-order neuro-fuzzy inference systems with efficient training, which is relevant when decision transparency is required in governance-oriented applications (Mohammadzadeh et al., 2025). The model has proved quite successful in NLP, where linguistic tasks tend to overlap semantically or structurally.

Recent work has shown the capability of multi-task learning to enhance performance on a range of NLP tasks. For example, certain frameworks integrate text style transfer and paraphrase generation into a single training framework (Bujnowski et al., 2020), while others utilize relevance-based sampling techniques to promote text generation across the tasks (Sai et al., 2022). These examples validate that related tasks can support one another when learned together, resulting in more generalizable and context-aware systems. Similar reinforcement principles have been employed in many-objective optimization to handle irregular decision variables (Xing et al., 2022a).

In this research, we apply a multi-task learning method to collaboratively model audit issue classification and audit opinion generation. A comparable synergy between task interdependence and optimization has been explored in cloud workflow planning for execution efficiency (Xing et al., 2022b). Joint learning approaches under uncertainty have also demonstrated effectiveness in logistics and maritime network optimization (Zhang et al., 2023). Both tasks are naturally correlated in actual audit processes, where auditors commonly examine issues and write audit opinions simultaneously. The practical co-occurrence suggests a hidden semantic correlation that can be learned jointly, finally making the two tasks more accurate and coherent.

3. Audit issue classification and audit opinion generation for electric power system

3.1. Audit issue classification

Text categorization is a core problem in NLP and has found extensive use in many domains of research. For the case of electric power auditing, a number of prominent items of data are normally written in natural language, e.g., audit description, provision referred to, and opinion of audit, as shown in Table 1. Among these, the audit description is manually written by auditors to document the audit process and its findings. This component plays a critical role in supporting decision-making and regulatory oversight within electric power enterprises.

To organize and manage audit records effectively, each entry must be assigned a corresponding issue category. This labeling process, referred to in this study as audit issue classification, is traditionally performed manually. However, manual labeling at scale is labor-intensive, time-consuming, and susceptible to human error. As audit datasets continue to grow in size and complexity, there is a pressing need for automated and efficient classification techniques that can accurately assign categories to audit issues without compromising reliability.

In this work, audit issue classification is framed as a supervised learning problem, where the objective is to learn a conditional probability distribution $p(c \mid x)$. Here, c denotes the category label, and x represents the input data. Typically, x is a single piece of text, but in our case, the classification decision must be informed by multiple input elements. In particular, we take both the audit description and the referenced provision to be crucial in deciding on the proper category. This can be seen from the examples of Table 1, where the problem category is very close to both textual elements.

To make the problem formal, let d be the audit description and r the referred provision. The task of classification can thus be formulated as learning the conditional distribution $p(c \mid d,r)$. By considering these two input features together, the model is in a better position to learn the semantics and context required for correct classification. This formulation allows for a more holistic interpretation of the audit record and facilitates the construction of robust, scalable classification solutions for electric power audit systems.

3.2. Audit opinion generation

In typical electric power auditing processes today, audit opinions are normally written manually by the auditors. Yet, as noted in the examples in Table 1, audit opinions actually encapsulate well the audit description and the cited provision. This tight semantic coordination implies that the audit opinions could be automatically produced using NLG technology. Unlike classification, which assigns a discrete label, the goal of audit opinion generation is to produce a coherent sequence of words based on input content. Formally, this task aims to maximize the following likelihood function:

$$p(y|d,r) = \prod_{t=1}^{T} p(y_t|y_{<t},\ d,\ r)\ ; \qquad y_{<t} = (y_1,\ldots,y_{t-1}) \tag{1}$$

In this equation, y_t denotes the t-th token in the output sequence, and x is the input composed of the audit description d and the referenced provision r. The generation model must learn the conditional distribution $p(y \mid d,r)$, where y represents the full audit opinion.

3.3. Multi-task learning framework for audit issue classification and audit opinion generation

While audit issue classification and audit opinion generation can be studied as separate tasks, treating them independently limits their effectiveness. These tasks are semantically related and often completed together in practical auditing. This interdependence allows for mutual enhancement, where progress in one task can support and inform the other. Therefore, we propose a multi-task learning approach to optimize both tasks simultaneously. Here, we jointly learn two task likelihoods, $p(c \mid d,r)$ for classification and $p(y \mid d,r)$ for generation, under shared parameters r induced by the dual encoders and the fused representation H. During training, both tasks contribute gradients through a joint loss to shape H; during inference, each task produces its output from H without conditioning on the other task's predicted label or generated text. The benefit of multi-task learning comes from shared representation learning under complementary supervision. The classification objective encourages H to preserve discriminative cues that separate standardized issue categories, while the generation objective encourages H to preserve fine-grained semantic relations needed for provision-consistent and coherent opinions. Joint optimization, therefore, serves as an inductive bias that improves generalization for both tasks, which is reflected by the observed metric changes when removing either auxiliary task in the ablation results.

To achieve this, we create a multi-task learning architecture as described in Fig. 1. The input has two sequences: (i) the audit description $d = \{d_1, d_2, \ldots, d_m\}$ and (ii) the referenced provision $r = \{r_1, r_2, \ldots, r_n\}$, where m and n are the lengths of both sequences. We employ two encoders for obtaining contextual representations for each of the inputs. For the description of audit d, we use a pre-trained BART encoder (Kelley and Hagan, 2025), which is best suited for natural language sequences. For the cited provision r, which normally involves technical legal

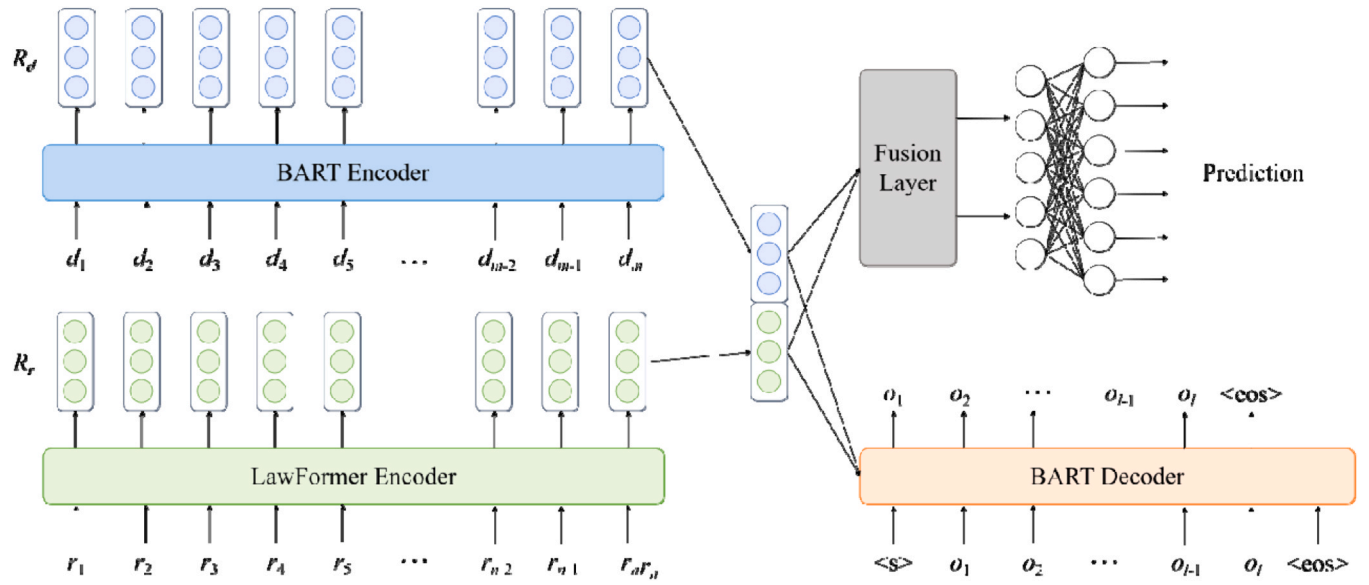


Fig. 1. Proposed multi-task framework with dual encoders for audit descriptions and referenced provisions, fusion via self-attention, and two outputs for joint issue classification and audit opinion generation.

language, we use LawFormer (Xiao et al., 2021), a language model pre-trained only on legal documents in order to more effectively comprehend such content. This dual-encoder design is motivated by the heterogeneous distributions of audit text: the audit description is narrative and context-dependent, whereas the referenced provision is normative and often contains legal or policy-style long-document patterns. Using specialized pre-trained encoders reduces representation mismatch and improves domain coverage, which is difficult to achieve with a single shared encoder under limited labeled audit data.

$$R_d = \text{BART}(d) = \{R_{d1}, R_{d2}, \cdots, R_{dm}\} \tag{2}$$

$$R_r = \text{LawFormer}(r) = \{R_{r1}, R_{r2}, \cdots, R_{rn}\} \tag{3}$$

Both BART and LawFormer use self-attention mechanisms to encode the input text, producing high-dimensional vector representations for each token. The resulting matrices R_d is $m \times d_v$ and R_r is $n \times d_v$ represent the contextual embeddings for the audit description and referenced provision, respectively, where d_v is the hidden dimension size (set according to the selected pretrained backbone, e.g., 768 for the base variants used in our experiments).

To enable interaction between the two input components, we design a fusion module. This module combines R_d and R_r and applies a self-attention mechanism to model cross-dependencies. The fusion layer functions as an explicit semantic alignment module between "fact descriptions" and "normative criteria". By applying self-attention on the concatenated token representations, the model learns which provision fragments are most relevant to particular description fragments, thereby producing a shared representation H that is simultaneously discriminative for category prediction and informative for coherent, provision-consistent opinion generation. The first step is to concatenate the two matrices into a single representation R:

$$R = \text{concatenate}(R_d, R_r) \tag{4}$$

Following the method described in (Sai et al., 2022), we introduce three parameter matrices W^Q, W^K, and W^V and compute the self-attention output as:

$$H = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_v}}\right)V \tag{5}$$

Here, H denotes the resulting hidden representation, and Q, K, and V are derived by projecting the input R through the respective weight matrices:

$$\begin{cases} Q = W^Q R \\ K = W^K R \\ V = W^V R \end{cases} \tag{6}$$

The output H serves as a shared feature representation used for both classification and generation tasks. For audit issue classification, we first apply a max-pooling operation to H, followed by a fully connected layer D_1 with a Sigmoid activation, and then another fully connected layer D_2 with a Softmax function to predict the final category label:

$$H_1 = \sigma(D_1(\text{MaxPool}(H))) \tag{7}$$

$$\text{Out} = \text{Softmax}(D_2(H_1)) \tag{8}$$

Here, $\sigma(\bullet)$ denotes the Sigmoid activation function. Here, Sigmoid is used only as a hidden-layer activation, whereas the final category prediction is obtained via Softmax over all classes for single-label multi-class classification. For audit opinion generation, we adopt a self-regressive decoding approach, where output tokens are generated sequentially using a Transformer decoder until a designated end-of-sequence token ("<eos>") is produced, as illustrated in Fig. 1.

4. Experiments

4.1. Data

We constructed a dedicated dataset based on internal audit records obtained from an electric power enterprise, comprising approximately 9500 entries. Each entry contains multiple structured and unstructured fields, including the audit description, audit issue category, audit opinion, referenced provision, monetary amount involved, and audit department. For the purpose of this study, we selected the two most informative fields: audit description and referenced provision, as input features. The model then predicts the audit issue category and generates the corresponding audit opinion as output.

Since the data is in Chinese, we remove uncommon or noisy non-text symbols while preserving Chinese characters, Arabic numerals, common monetary and percentage symbols (for example, 'CNY' and '%'), and standard punctuation. In tokenization, we adhered to the segmentation scheme employed in the Chinese adaptation of the BART model (Wu et al., 2025), which does text segmentation at the character level and not

by whole words. This provides consistency throughout all models by using each Chinese character or punctuation mark as a single token embedding.

The data set was split into training, validation, and test sets in the ratio of 8:1:1. The split is stratified by audit issue category to preserve the class distribution across training, validation, and test sets. In addition, we remove exact duplicate records and ensure that identical audit description text does not appear in more than one split, which reduces near-duplicate leakage in evaluation. The training set is employed for tuning the model parameters, and the validation set is employed to track performance after every epoch in order to enable early stopping in order to avoid overfitting and improve generalization. The ultimate evaluation is performed on the test set using various measures to evaluate classification accuracy and generation quality.

4.2. Implementation details

All experiments were performed on a high-performance internal Graphics Processing Unit (GPU) cluster. The hardware setup included sixteen Intel Xeon Silver 4114 Central Processing Unit (CPUs) operating at 2.20 GHz and two NVIDIA Titan V GPUs, providing a total of 24 GB of GPU memory. The neural network architecture, including the multi-task learning framework, was implemented using PyTorch and the Transformers library.

For optimization, we used AdamW with learning rate 3×10^{-5}, weight decay 0.01, and gradient clipping at 1.0. The maximum input length was 512 tokens for both d and r. We used a linear learning-rate scheduler with a 10% warmup ratio. The batch size for all experiments was fixed at 8. Both the audit issue classification loss L_C and the audit opinion generation loss L_G were defined using cross-entropy loss functions. These were combined into a unified loss function L, controlled by a weighting hyperparameter α, to measure the overall discrepancy between predicted and actual outputs:

$$L = \alpha \cdot L_C + (1 - \alpha) \cdot L_G \tag{9}$$

Initially, the value of α was set to 0.5. To determine the optimal setting, we adjusted α in increments of 0.1 across multiple experiments. Based on the results, the optimal value was found to be 0.4. We selected α by comparing validation performance across both tasks, using macro-F1 for classification and ROUGE-L for generation as the primary indicators to identify the best overall tradeoff. To further justify the choice of α and assess robustness, a sensitivity analysis over $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$ is conducted, using macro-F1 for classification and ROUGE-L for generation. The reported results quantify the tradeoff between the two tasks and provide practical guidance for setting α under different application priorities.

We applied an early stopping strategy during training to avoid overfitting. After each training epoch, the model's performance was evaluated on the validation set. If the validation loss stopped decreasing, training was halted. The final model was then assessed using the test set. The complete training process consisted of 17 epochs.

4.3. Evaluation metrics

To evaluate the performance of our multi-task learning framework, we design two sets of evaluation metrics: one for audit issue classification and the other for audit opinion generation.

Audit issue classification is formulated as a standard multi-class classification task. Therefore, we adopt four widely used metrics: Accuracy, Precision, Recall, and F1 Score. Precision, Recall, and F1 Score are reported as macro-averages across all classes, while Accuracy is computed over all test instances. These metrics provide a comprehensive assessment of classification performance by quantifying the proportion of correct predictions, the correctness of positive predictions, the coverage of actual positives, and the harmonic balance between precision and recall, respectively. Readers seeking detailed definitions and formulas for these metrics may refer to (Sai et al., 2022).

For audit opinion generation, we adopt a set of established evaluation metrics commonly used in NLG tasks: BLEU, ROUGE, Distinct-n (DISTINCT), and BERTScore. Each metric captures a different aspect of generation quality:

- BLEU-n evaluates the degree of n-gram overlap between the generated text and the reference text. It reflects fluency and precision in word choice.
- ROUGE-n measures the n-gram recall rate between the generated output and the ground truth. ROUGE-L, a variant based on the longest common subsequence, is also used to better capture structural similarity.
- DISTINCT-n assesses the diversity of the generated text by computing the ratio of unique n-grams in the output, which helps prevent repetitive or generic responses.
- BERTScore evaluates semantic similarity by comparing contextual embedding of the generated and reference texts using a pre-trained BERT model. This metric captures meaning beyond surface-level lexical overlap.

All BLEU and ROUGE scores are computed at the character level, consistent with our Chinese tokenization, where each Chinese character and standard punctuation mark is treated as one token.

Following standard practice in prior work, we set n = 1,2,3,4 for BLEU, adopt ROUGE-1, ROUGE-2, and ROUGE-L, and apply DISTINCT-1 and DISTINCT-2 to measure diversity (Sai et al., 2022). For clarity, we report ROUGE-1, ROUGE-2, and ROUGE-L in Table 3, consistent with the metric definitions in this section.

4.4. Experimental results

4.4.1. Audit issue classification results

Table 2 and Table 3 present the detailed results for both tasks. For the audit issue classification task, we compare our proposed model against a range of baseline methods. These include traditional machine learning models such as Naive Bayes, Support Vector Machines (SVM), Gradient Boosting Decision Trees (GBDT), AdaBoost, and XGBoost. In addition, we evaluate deep learning models commonly used in text classification, including TextCNN and LSTM, as well as the pre-trained language model BERT.

For traditional machine learning baselines, we represent the audit description and referenced provision using the bag-of-words approach. For the deep learning models TextCNN and LSTM, we concatenate the audit description and referenced provision, then use Word2Vec to obtain their vector representations before training the models in an end-to-end fashion. In the case of BERT, we also concatenate the two inputs and apply Byte Pair Encoding (BPE) tokenization along with its pre-trained

Table 2
Audit issue classification performance comparison in terms of Accuracy and macro-averaged Precision, Recall, and F1 across baselines, ablations, and the proposed model.

Algorithm	Accuracy	Precision	Recall	F1-score
NB	0.6042	0.6217	0.595	0.6081
SVM	0.6203	0.6329	0.63	0.6314
GBDT	0.6415	0.6532	0.6628	0.6579
AdaBoost	0.6636	0.6577	0.6698	0.664
XGBoost	0.667	0.6672	0.6704	0.6688
TextCNN	0.7165	0.7427	0.6901	0.7156
LSTM	0.7278	0.7439	0.705	0.7239
BERT	0.7791	0.7823	0.7794	0.7808
BART	0.7803	0.7819	0.7799	0.7809
Proposed	**0.8273**	**0.8331**	**0.8288**	**0.8309**
w/o. d	0.7458	0.7521	0.762	0.757
w/o. r	0.7882	0.7799	0.7805	0.7802
w/o. Gen	0.8051	0.7996	0.8088	0.8042

Table 3
Evaluation results for audit opinion generation across multiple metrics, highlighting the comparative effectiveness of the proposed multi-task learning framework.

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	Dist-1	Dist-2	BERT-Score
Seq2Seq	32.76	30.18	27.44	23.09	0.413	0.165	0.425	**0.458**	0.142	0.794
GPT-2	40.35	37.84	32.23	29.08	0.459	0.296	0.466	0.317	0.128	0.845
BART	41.64	39.33	36.58	33.3	0.478	0.332	0.472	0.303	0.116	0.859
Our	**47.09**	**45.27**	**42.69**	39.9	**0.522**	**0.389**	**0.513**	0.289	0.109	**0.877**
w/o. *d*	42.31	39.86	36.9	32.81	0.469	0.334	0.473	0.261	0.116	0.862
w/o. *r*	42.07	39.65	36.47	33.04	0.454	0.326	0.451	0.244	0.104	0.855
w/o. CLS	46.61	44.39	42.07	**40.11**	0.504	0.383	0.479	0.288	**0.155**	0.869

embeddings.

As shown in Table 2, our proposed multi-task learning framework achieves superior results across all four evaluation metrics compared to the baselines. Notably, the F1 score improves by approximately five percentage points over both BERT and the BART encoder, representing a significant performance gain.

To further investigate the design of our framework, we conduct a series of ablation experiments. First, we evaluate the impact of each input component by selectively removing the audit description and referenced provision. These modified configurations are recorded as "w/o. *d*" and "w/o. *r*" in Table 2, where "w/o." denotes "without". When either input is excluded, the fusion layer becomes redundant, and the resulting hidden vectors are used directly for classification and opinion generation. The results show that removing either input leads to a drop in performance across all metrics. In particular, omitting the audit description results in a more substantial decline, with the F1 score decreasing by approximately 7.3 %age points. When only the referenced provision is removed, the drop is around five percentage points. These findings confirm that while both inputs contribute useful information, the audit description plays a more critical role in classification.

To evaluate the benefit of multi-task learning itself, we also test a single-task configuration by removing the BART decoder. In this case, the model retains only the fully connected prediction layer for classification, and the loss function for audit opinion generation is set to zero. This variant is marked as "w/o. Gen" in Table 2. Accordingly, "w/o. Gen" serves as an architecture-matched single-task classification baseline, and the performance gap to the full model reflects the benefit brought by auxiliary generation supervision. The results indicate a slight decrease in performance, with the F1 score dropping from 0.8309 to 0.8042. This demonstrates that including the opinion generation task during training helps the model learn more effective representations for audit issue classification, thereby validating the advantage of multi-task learning.

4.4.2. Audit opinion generation results

To evaluate the effectiveness of our proposed model in generating audit opinions, we compare its performance against three baseline methods:

1) Seq2Seq (Wu et al., 2025), a foundational neural sequence generation model introduced in 2014. This model uses a Recurrent Neural Network (RNN) architecture for both encoder and decoder and does not rely on pre-trained parameters.

2) GPT-2, a Transformer decoder pre-trained on large-scale general-domain corpora, widely used in open-ended text generation tasks.

3) BART (Soper et al., 2021), a pre-trained Transformer-based encoder-decoder model, designed for sequence generation tasks using denoising objectives on extensive corpora.

For each of these baseline models, the input consists of a concatenated string comprising the audit description and referenced provision, separated by a special token "[SEP]" (Separator Token). The output is the generated audit opinion.

In contrast, our proposed method treats the audit description and referenced provision as separate inputs. Each is processed by a dedicated encoder: BART for the audit description and LawFormer for the referenced provision. These encoded representations are then fused using a self-attention-based fusion layer to extract mutual dependencies. Furthermore, the architecture integrates audit issue classification as an auxiliary task within a multi-task learning framework, enabling shared learning signals to enhance audit opinion generation.

Table 3 presents the evaluation results. From the experimental results, several key observations can be made.

- First, in terms of BLEU and ROUGE metrics, our model consistently outperforms all baseline methods. This demonstrates that the audit opinions generated by our framework exhibit stronger alignment with human-written ground-truth outputs, both in terms of lexical similarity and structural coherence.
- Second, while our method achieves slightly lower DISTINCT scores than BART and considerably lower scores than Seq2Seq and GPT-2, this result is consistent with expectations in the audit domain. Unlike open-domain generation tasks where high DISTINCT values indicate desirable diversity, audit opinions typically follow standardized formats and stylistic conventions, even when authored by different individuals. As a result, lower DISTINCT values in this context suggest greater consistency and accuracy, rather than a limitation in output variability.
- Finally, our model also surpasses all baselines on the BERTScore metric, which measures semantic similarity. This confirms that the audit opinions generated by our model are not only lexically aligned but also semantically faithful to the original input, leading to higher overall quality in the generated content.

We also conduct ablation studies, shown in the lower part of Table 3, to analyze the contribution of each model component. When the BART encoder or the LawFormer encoder is removed, BLEU scores drop significantly, although the model still performs better than the strongest baseline, BART. ROUGE scores also decline, and small reductions are observed in both DISTINCT and BERTScore values. These findings demonstrate that both the audit description and the referenced provision play critical roles in generating accurate audit opinions.

Unlike audit issue classification, the removal of *d* and *r* leads to similar drops in performance. In this case, the configuration without *d* performs only slightly better than the configuration without *r*, suggesting that both input components are equally important for opinion generation.

Additionally, we test the model without the audit issue classification task (denoted as "w/o. CLS" in Table 3). The variant without the audit issue classification task shows mixed changes across metrics, including a slight increase in BLEU-4, while ROUGE and BERTScore decrease overall. This indicates that using classification as an auxiliary objective generally improves semantic alignment and coherence in the generated audit opinions. Because "w/o. CLS" retains the same dual-encoder and fusion structure but removes the auxiliary classification objective, it serves as a strong generation baseline. The performance gap between the full model and "w/o. CLS" therefore isolates the contribution of classification supervision to ROUGE-L and BERTScore, reflecting improved semantic alignment and provision consistency.

5. Conclusion

Auditing is a critical task within electric power enterprises, directly influencing operational transparency, regulatory compliance, and ultimately, enterprise revenue. In increasingly digitalized power and integrated energy operations, strengthening audit intelligence also supports resilience-oriented governance by improving compliance visibility and accelerating corrective actions under operational uncertainty. Among the most labor-intensive aspects of audit work are audit issue classification and audit opinion generation. While these two tasks have traditionally been approached separately, this paper demonstrates that they are semantically interdependent and can benefit from a joint modeling approach.

To address this, we proposed a novel multi-task learning framework that simultaneously performs audit issue classification and audit opinion generation. The model employs two distinct encoders to process the audit description and referenced provision, fuses the extracted features through a self-attention-based fusion layer, and generates both outputs through a classification head and a Transformer decoder. Experimental results on a real-world dataset validate the superiority of our method. Compared to single-task learning baselines, our framework improves classification F1 score from 0.8042 to 0.8309, and demonstrates notable gains in text generation quality with significant increases in BLEU and ROUGE metrics. In ablation studies, removing either input component resulted in performance drops of up to 7.3% points in classification and meaningful declines in generation quality, confirming the importance of both inputs.

Furthermore, we explored the influence of key hyperparameters and input configurations, offering practical recommendations for deploying intelligent audit systems in operational settings. The proposed framework not only enhances audit efficiency and accuracy but also provides a scalable solution adaptable to other auditing domains beyond electric power enterprises.

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CRediT authorship contribution statement

Qinglin Meng: Conceptualization, Writing – Original draft, Visualization. **Weiqiang Jin:** Methodology, Software. **Youchun Zhuang:** Data Curation. **Lei Guo:** Formal Analysis. **Jianbin Xu:** Resources. **Guanqiao Zhao:** Investigation. **Chao Ma:** Validation. **Ying He:** Validation. **Sheharyar Hussain:** Writing – Review & Editing. **Deyi Xiong:** Supervision, Project Administration, Funding Acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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